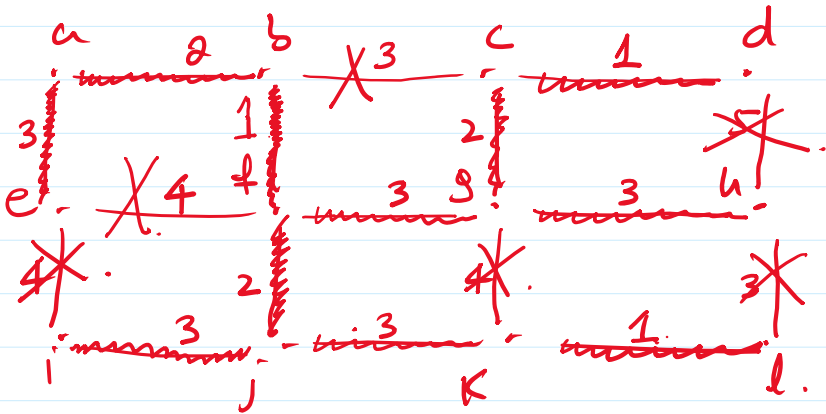


lecture 24.

Minimum SPANNING TREE.

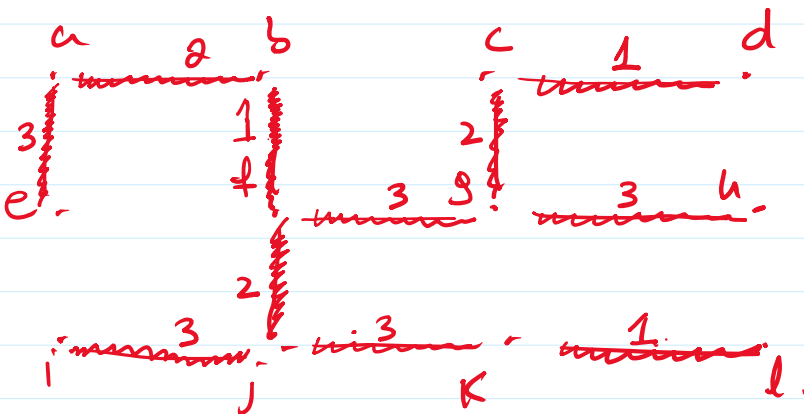
- Connected
- Undirected
- No Simple Circuit.
- Contains all Vertices.
- Subgraph.
- Sum of weights of edges is minimum.

PRIMS.

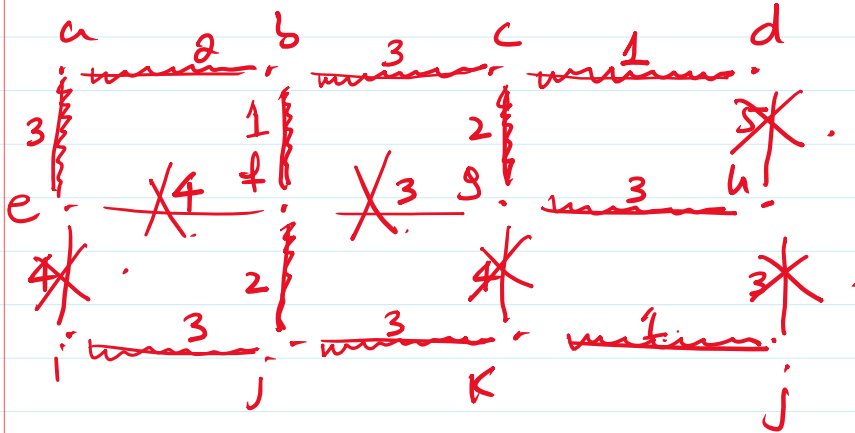


edge	weight.
(b, f)	1.
(a, b)	2.
(f, j)	2.
(a, e)	3.
(i, j)	3.
(f, g)	3.
(c, g)	2.
(c, d)	1.
(g, h)	3.
(j, k)	3.
(k, l)	1.

24.



KRUSKAL.

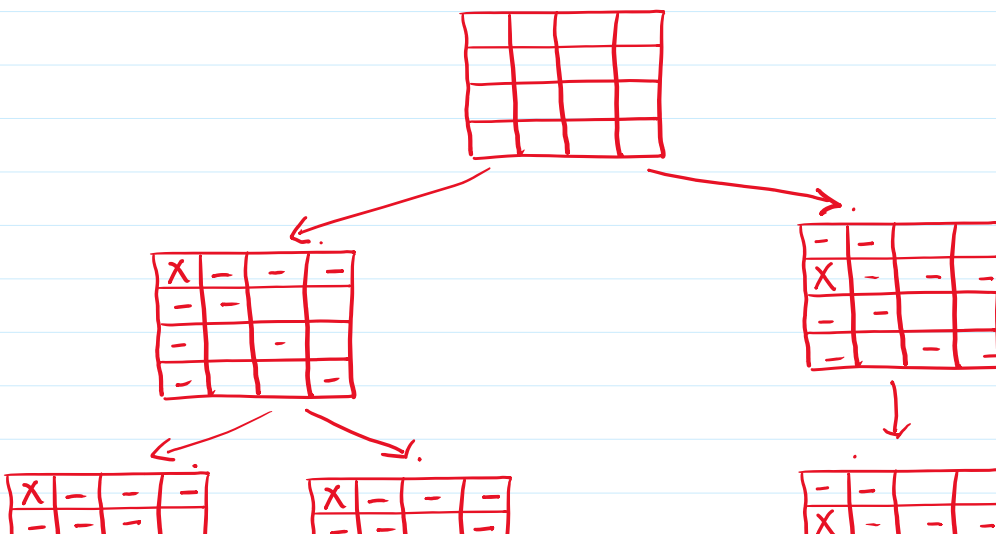


edge	weight.
(c,d)	1.
(k,l)	1.
(b,c)	1.
(a,g)	2.
(f,i)	2.
(a,b)	2.
(b,c)	3.
(j,k)	3.
(g,h)	3.
(i,j)	3.
(a,e)	3.
	24.

BACKTRACKING.

Ex 7669

1- Queen problem.



$$\begin{array}{|c|c|c|c|} \hline X & - & - & - \\ \hline - & - & - & - \\ \hline - & X & - & - \\ \hline - & - & - & - \\ \hline \end{array}$$


Dead end.

$$\begin{array}{|c|c|c|c|} \hline X & - & - & - \\ \hline - & - & - & - \\ \hline - & - & - & - \\ \hline - & X & - & - \\ \hline \end{array}$$


$$\begin{array}{|c|c|c|c|} \hline X & - & - & - \\ \hline - & - & X & - \\ \hline - & - & - & - \\ \hline - & X & - & - \\ \hline \end{array}$$


Dead end.

$$\begin{array}{|c|c|c|c|} \hline - & - & - & - \\ \hline X & - & - & - \\ \hline - & - & - & - \\ \hline - & X & - & - \\ \hline \end{array}$$


$$\begin{array}{|c|c|c|c|} \hline - & - & X & - \\ \hline X & - & - & - \\ \hline - & - & - & - \\ \hline - & X & - & - \\ \hline \end{array}$$

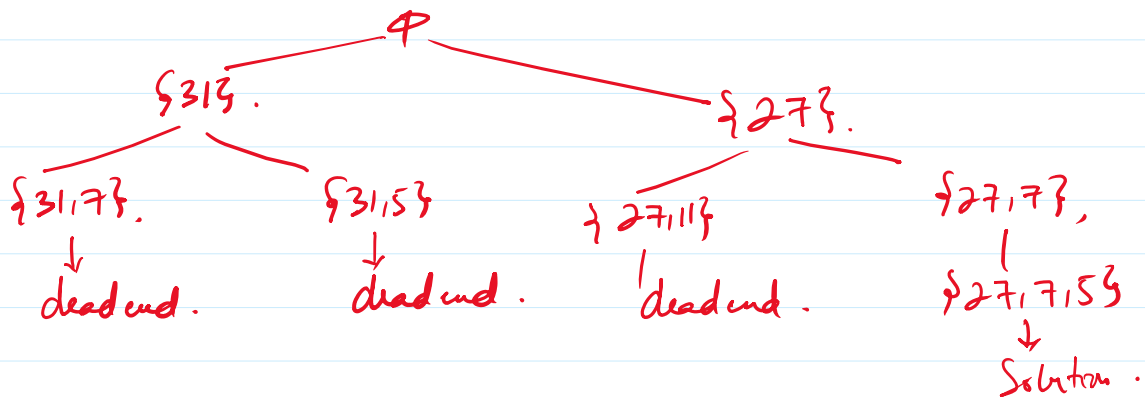

$$\begin{array}{|c|c|c|c|} \hline - & - & X & - \\ \hline X & - & - & - \\ \hline - & - & - & X \\ \hline - & X & - & - \\ \hline \end{array}$$


Solution.

Ex 8
671.

{31, 27, 15, 11, 7, 5}.

Subset Sum = 39.



Ex 32:-
674

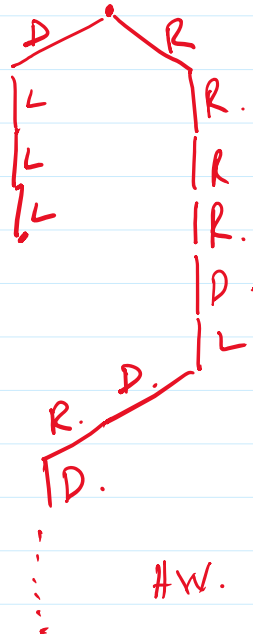
U, D, R, L.

$$\begin{array}{|c|c|c|c|c|c|c|c|} \hline . & . & . & . & . & . & . & . \\ \hline . & . & . & X & X & X & X & X \\ \hline X & X & X & X & . & . & X & X \\ \hline . & . & . & . & . & . & X & X \\ \hline . & . & . & . & . & . & . & X \\ \hline \end{array}$$


Exit.



Exit.



Final:-
50%.

Session 1 :- Quantifiers. (1).
5% ± 2%.

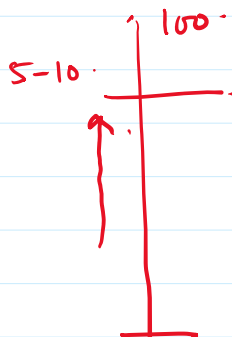
Session 2:- Relations representation
+ properties. 5% ± 2%.

Final :- PARTIAL ORDER - 40% ± 5%.

BACKTRACKING.

FEEDBACK.

- | | |
|-----------------|-------|
| 1 - Content. | ITEC. |
| 2 - Evaluation. | RAT. |
| 3 - Delivery. | ✓. |



80-90.

- 1) Break.
- 2) Practice.

5-10

[illegible]

COMPANY A

COMPANY B.

- Location.
- Revenue.
- Size.
- Survey.
- Effort.
- Promotion.
- Training.
-
-
-
-

100 .

WEB

So

Security

Network

2

5

3

4

6 m

100

12

1



Surveys.